BYO AI ACCOUNT BRIDGE / RESEARCH SERIES

Research & Feasibility

Technical possibility, competitive landscape, permission gates and evidence quality

01

DETAILED TECHNICAL DOSSIER

User-owned accounts / Local-first runtime / Optional cloud relay

Research cutoff: 26 October 2026

A Consumer-web adapter (permission-gated)

B Official account authorization where supported

C User-owned free APIs and locally running models

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> Research notation: [Sxx] markers link to first-party terms, official technical documentation, published standards or original project repositories. Statements labeled proposed, target, or estimate are design judgments, not observed performance.

Executive finding and research boundaries

Product thesis. Give each developer one OpenAI-style endpoint backed by AI capacity that the developer already controls: a consumer-web session when its provider authorizes automation (A), an approved account authorization flow (B), or user-owned free API quotas and local models (C). The product does not supply or pool third-party accounts and does not promise unlimited generation.

Research verdict. The transport, format translation, browser control and optional remote relay are technically achievable. The primary risk is not whether a script can read a chat response; it is whether any target service permits that kind of automated consumer access, whether the integration survives product changes, and whether the user understands the limitations. Competing open-source projects demonstrate implementations but not broad permission to commercialize them. **[S16] [S17] [S18]**

> KEY DECISION: Build a local-first, provider-agnostic bridge. Gate every A connector on documented permission. Prioritize an authorized B connector and C providers so the MVP remains useful even if no major proprietary chat website approves A.

The evidence standard in this series is: **confirmed** means the provider or standards body documents a capability or rule; **observed project claim** means an original repository says it implements something; **proposed** means our engineering design; **unknown** means it must be experimentally tested or negotiated. This avoids treating reverse-engineered code as proof of supported access.

What the idea is, and what it is not

Three routes to the same user-facing contract

Route	Actual source of model inference	Distinguishing constraint
A: Consumer web	User's signed-in chat interface, when automated use is authorized	Web-session permission, fragile UI, consumer quotas
B: Official account grant	Provider's supported authorized model route billed against eligible account allowance	Provider-specific eligibility and supported operations
C: Free API / local	Official free-tier API key or a locally hosted open-weight model	Quotas or device compute resources

The gateway exposes API-like responses; MCP is an optional tool transport and does not itself grant free model access. The July 2026 MCP specification deprecated sampling, so a new product should not rely on borrowing a client's model through sampling as its core engine. **[S14]**

Not an AI reseller. Each person controls their own accounts and permissions. The hosted service may sell dashboard conveniences later, but never silently route one customer's prompt through another person's credentials. **Not an unlimited service.** Requests stop or move only to explicitly authorized alternatives when a quota is exhausted.

Current provider permission and product landscape

Consumer access versus programmable access

Service	Direct consumer-web A assessment	Supported alternative and implication
OpenAI ChatGPT	Individual terms prohibit automated/programmatic extraction and bypassing limits **[S01]**	ChatGPT plan-usage OAuth is documented for eligible local/open-source tools; API limits and support differ **[S04] [S05]**
Anthropic Claude	Consumer terms restrict non-human access without API authorization/explicit permission; commercial consumer credential routing is restricted **[S02] [S03]**	Own Anthropic API key, or unmodified official Claude Code under the terms **[S03]**
Google Gemini	General Google terms restrict abusive/unauthorized automated access; direct site permission must be separately verified **[S24]**	Gemini API has a genuine free tier on eligible models/projects, with data and regional conditions **[S08] [S09]**
Open-weight local	No consumer website session to automate	Ollama already provides OpenAI-compatible local inference **[S13]**

Subtle but important: The fact that a user owns an account does not override that provider's agreement about automation. A local-only product reduces centralized credential custody but does not, by itself, turn a prohibited scrape into authorized access. A connector may therefore be technically ready yet disabled for release.

Important new authorized avenue: ChatGPT plan usage

OpenAI documents a Sign in with ChatGPT flow that can grant eligible locally hosted open-source applications account-plan usage for supported Responses API requests. OAuth authorization is separate from simple identity sign-in; the user approves plan usage and can manage limits. The documented inference surface is the public `/v1/responses` route, not undocumented ChatGPT website endpoints. The current preview requires `store:false`, `stream:true`, complete explicit input context, and excludes multiple hosted tools and fields. **[S04] [S05] [S06] [S32]**

This is a promising implementation of **B**, not automatic permission for **A**. Commercial or remotely hosted integrations may require provider approval. Eligibility differs by account and app. Public help pages describe plan sharing and usage controls for participating apps; we should display actual grant status rather than assume any free account can invoke inference. **[S07] [S34]**

Competitive landscape and differentiation

Original project / product	What is relevant	Gap or caution for this proposal
Universal Web API [S16]	Local browser-driven proxy for ChatGPT, Claude, Gemini and others, with standard API facades	Warns against high-frequency and commercial use; similar A concept exists
WebAI-to-API [S17]	Gemini web/browser adapters, OpenAI-compatible endpoints, streaming, stateless chat variants	README warns of missing caller API authentication and feature differences
CLIProxyAPI [S18]	Unifies multiple CLI/account channels behind familiar model APIs	Provider-specific terms remain; broad account proxying is not proof of authorization
LiteLLM [S15]	Unified AI gateway, routing, logging, fallbacks, budgets and MCP	Mature gateway competition; generic routing alone is not differentiation
OpenRouter [S12]	Already aggregates models and offers free-model API access	Users can call free models directly; value must exceed mere key forwarding
Ollama [S13]	Local OpenAI-compatible model server	Running models locally is already straightforward for technical users

Proposed differentiation: a simple user-account connection experience, honest per-connector capability labels, user-owned credential boundaries, a local/remote switch with explicit privacy warnings, precise quota handling, and free-first use cases for coding agents and student projects. Avoid an unsupported claim such as "all premium AI models for free."

Competitive proof is not market proof

Existing repositories are evidence of developer implementation interest, not evidence of an addressable market, retention, willingness to pay, conversion or uptime. Stars, README feature claims and social commentary are biased convenience signals. No independent market-size estimate is presented as a verified fact in this research.

Technical feasibility by subsystem

Subsystem	Feasibility	Main blocker / proof required
Local OpenAI-style REST facade	High	Request/response conformance, auth, SSE cancellation
Browser-session adapter framework	High as a framework	Per-site authorization and resilience to UI changes
Website system prompts and function calls	Partial	Consumer UI may not expose true roles or structured tool calls
Official authorized account connector	High where eligible	Scopes, token refresh, preview restrictions **[S05] [S06]**
Free API connectors	High	User-specific rate limits, region, free eligibility **[S09] [S10] [S35]**
Local Ollama connector	High	Hardware speed and memory; model quality/capability **[S13] [S36]**
Online dashboard	High	Device registration, access controls, privacy choices
Remote API relay	High with engineering effort	Offline devices, WSS authentication, backpressure, data exposure **[S19]**
Full model API feature parity	Low for A	Native tools, exact token usage, model selection, audio/files

Browser prototype reality check

Browser automation can operate a signed-in website under user control, and Playwright supports isolated authenticated contexts. That only establishes a tooling mechanism. UI response extraction may return partial text; "streaming" can be an evolving page rather than structured token events; web chat may hold implicit conversation state; model names, token counts and error conditions may be inaccessible. **[S21]**

The feasibility test must therefore separate: **can send**, **can reliably read**, **can normalize**, **can legally distribute**, and **can maintain**. Do not merge those into a single "works" verdict.

Unit economics and operating assumptions

Our venture does not buy model tokens for user-owned B/C accounts by default. It still incurs dashboard, auth, database, device-connectivity, bandwidth, support and software maintenance costs. Local A and local Ollama use the user's device and network. Remote relay requests transit the platform, creating operational and privacy responsibilities.

Illustrative sizing, not a vendor quote: 100 users x 10 requests/day = 1,000 requests/day. If an average request+response payload were 12 KiB, traffic carrying just those payloads would be about 12 MiB/day, before TLS framing, retries, metadata, SSE overhead, monitoring and software downloads. Large image/file requests change the picture drastically. Long-lived tunnels, support and

implementation work can dominate this example; measure real bytes and connection-minutes in beta.

One provider's free quota is not a service capacity guarantee for a thousand requests; users must own their own entitlements, and fallbacks may be unavailable. OpenRouter publicly limits free model traffic, Groq publishes per-organization limits, and Gemini limits depend on project/model eligibility. Do not aggregate fake accounts to evade quotas. **[S10] [S11] [S12]**

Preliminary business options

- Open-source local engine, free forever for core self-use; preserves mission and earns trust.
- Optional paid convenience for dashboards, device monitoring, managed updates, or usage analytics - only after users validate value.
- Team features, organization policy, and private deployment as a later product, with provider legal review.
- Donations/sponsorships while early traction is unproven.

No revenue model should be based on reselling consumer chat allowance without express provider authority.

Risk ranking and evidence gaps

Risk	Likelihood / impact (judgment)	Mitigation or gate
Consumer automation rejected by providers	High / critical	Permission matrix; never promise unapproved connectors
Session breakage and changing UI	High / high	Contract tests, pinned adapters, feature flags, rapid disable
Credentials leak into cloud	Medium / critical	Local storage, schema-bound telemetry, security review
Cross-user remote request routing	Medium / critical	Device/user scoping, authorization on every call
Hidden or inaccurate model attribution	Medium / high	Expose "unknown/selected via UI" truthfully
Quotas reduce utility for free users	High / medium	Transparent status, permitted C options, local models
Competitors provide same workflow	High / medium	Research onboarding/UX advantage with users
Mobile or device-offline expectations	High / medium	Declare always-on local agent requirement

Research program and go/no-go decisions

Thirty-day evidence plan

- **Days 1-5:** version a permission matrix and source register for 3 candidate consumer interfaces; seek written clarification where terms are ambiguous. Build a mock chat UI for the A contract.

- **Days 6-10:** benchmark 50 authorized mock/real requests; measure completion, partial response detection, latency and logout recovery. Record no user secrets.
- **Days 11-15:** validate one official account grant, one free API and Ollama behind the same API contract. Test data-use notices.
- **Days 16-20:** interview 15 students/indie developers about existing spend, task mix, setup friction and remote requirements; distinguish stated interest from willingness to install.
- **Days 21-30:** run a supervised local pilot with 5 users; compare to directly configuring existing tools; decide whether hybrid cloud relay is justified.

Stage-gate questions

- G0 **Permission:** Is at least one meaningful A provider actually authorized for automated consumer access? If not, A remains a lab adapter rather than a public claim.
- G1 **Reliability:** Does supported text-only generation meet the agreed success target under quota without fabricating completions?
- G2 **Value:** Can non-expert developers reach first response materially faster than with existing gateways?
- G3 **Trust:** Can independent reviewers confirm consumer credentials never reach cloud storage or logs?
- G4 **Demand:** Are pilot users returning voluntarily after the first test?

Executive recommendation

Proceed with a **conditional A-first R&D program and a production-eligible B/C foundation**. Treat the A infrastructure as valuable engineering IP but no guarantee that major proprietary chat providers will allow it. The company's product promise should be "make your own available AI access usable" rather than "get paid models free." The next five volumes define the actual product, implementation, controls and delivery gates.

Evidence & source register

Sources were checked in October 2026. Their content, access conditions, prices and quotas can change. Project READMEs are evidence of what authors claim to implement, not proof of provider permission or reliability.

[S01] OpenAI consumer Terms of Use (effective January 2026)
https://openai.com/policies/row-terms-of-use/

[S02] Anthropic Consumer Terms of Service (effective October 2025)
https://cdn.sanity.io/files/4zrzovbb/website/43c13f51a6d9755341433f79cf63c966ed9200ea.pdf

[S03] Anthropic Claude Code - Legal and compliance
https://code.claude.com/docs/en/legal-and-compliance

[S04] OpenAI Sign in with ChatGPT - Local open-source plan usage
https://developers.openai.com/siwc/token-sharing-open-source

[S05] OpenAI Sign in with ChatGPT - Preview limitations
https://developers.openai.com/siwc/token-sharing-open-source/preview-limitations

[S06] OpenAI Sign in with ChatGPT - Models and inference
https://developers.openai.com/siwc/token-sharing-open-source/models-and-inference

[S07] OpenAI Help - Using ChatGPT plans in other apps
https://help.openai.com/en/articles/20001542-using-your-chatgpt-plan-in-other-apps-and-sites

[S08] Google Gemini API Additional Terms (effective March 2026)
https://ai.google.dev/gemini-api/terms

[S09] Google Gemini API billing and free tier
https://ai.google.dev/gemini-api/docs/billing

[S10] Google Gemini API rate limits
https://ai.google.dev/gemini-api/docs/rate-limits

[S11] GroqCloud API rate limits
https://console.groq.com/docs/rate-limits

[S12] OpenRouter pricing and free-model limitations
https://openrouter.ai/pricing

[S13] Ollama OpenAI-compatible API
https://docs.ollama.com/api/openai-compatibility

[S14] MCP specification release, 28 July 2026
https://blog.modelcontextprotocol.io/posts/2026-07-28/

[S15] LiteLLM AI Gateway official documentation
https://docs.litellm.ai/docs/simple_proxy

[S16] Universal Web API project README
https://github.com/lumingya/universal-web-api/blob/main/README.en.md

[S17] WebAI-to-API project README and limits
https://github.com/Amm1rr/WebAI-to-API

[S18] CLIProxyAPI project README
https://github.com/router-for-me/CLIProxyAPI

[S19] OWASP WebSocket Security Cheat Sheet
https://cheatsheetseries.owasp.org/cheatsheets/WebSocket_Security_Cheat_Sheet.html

[S21] Playwright authentication and browser-state guidance
https://playwright.dev/docs/auth

[S24] Google Terms of Service (July 2026)
https://policies.google.com/terms?hl=en-US

[S32] OpenAI - Sign in with ChatGPT account authorization
https://developers.openai.com/siwc/token-sharing-open-source/sign-in

[S34] OpenAI - Open-source sign-in integration guide
https://developers.openai.com/cookbook/articles/sign-in-with-chatgpt

[S35] Google Gemini API troubleshooting and eligibility
https://ai.google.dev/gemini-api/docs/troubleshooting

[S36] Ollama local quickstart
https://docs.ollama.com/quickstart

> Evidence boundary: this series is a design-and-research package, not a working implementation, legal opinion, measured security certification or promise of free access to any specific proprietary model.